SAMPLE

The Grand Unified Theory

The Quest to Unite All Forces

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Introduction

A Grand Unified Theory (GUT) is any model in particle physics that merges the electromagnetic, weak, and strong forces (the three gauge interactions of the Standard Model) into a single force at high energies. Although this unified force has not been directly observed, many GUT models theorize its existence.

If the unification of these three interactions is possible, it raises the possibility that there was a grand unification epoch in the very early universe in which these three fundamental interactions were not yet distinct.

Experiments have confirmed that at high energy, the electromagnetic interaction and weak interaction unify into a single combined electroweak interaction. GUT models predict that at even higher energy, the strong and electroweak interactions will unify into one electronuclear interaction. This interaction is characterized by one larger gauge symmetry and thus several force carriers, but one unified coupling constant. Unifying gravity with the electronuclear interaction would provide a more comprehensive theory of everything (TOE) rather than a Grand Unified Theory.

Thus, GUTs are often seen as an intermediate step towards a TOE. The novel particles predicted by GUT models are expected to have extremely high masses--around the GUT scale of 1016 GeV/c2 (only three orders of magnitude below the Planck scale of 1019 GeV/c2)--and so are well beyond the reach of any foreseen particle hadron collider experiments.

Therefore, the particles predicted by GUT models will be unable to be observed directly, and instead the effects of grand unification might be detected through indirect observations of the following: proton decay, neutron-antineutron oscillations, electric dipole moments of elementary particles, or the properties of neutrinos. Some GUTs, such as the Pati-Salam model, predict the existence of magnetic monopoles.

To see what such a theory is trying to do, imagine unwinding the history of the universe backwards. In today's cold, quiet cosmos, electricity and magnetism, weak radioactivity, and the strong nuclear force look like three unrelated interactions, each with its own carriers, its own strength, and its own book of rules. A Grand Unified Theory is the claim that this separateness is an illusion of low energy -- that in the furnace of the very early universe, the three were a single interaction, and that the machinery of symmetry breaking split them apart as the universe expanded and cooled. The same mathematics that describes how the electroweak force froze into electromagnetism and the weak force is now asked to finish the job.

Here, the unification of matter is even more complete, since the irreducible spinor representation 16 contains both the 5 and 10 of SU(5) and a right-handed neutrino, and thus the complete particle content of one generation of the extended Standard Model with neutrino masses.

Chapter 1: The Discovery of The Grand Unified Theory

Models that do not unify the three interactions using one simple group as the gauge symmetry but do so using semisimple groups can exhibit similar properties and are sometimes referred to as Grand Unified Theories as well. Historically, the first true GUT, which was based on the simple Lie group SU(5), was proposed by Howard Georgi and Sheldon Glashow in 1974.

The Georgi-Glashow model was preceded by the semisimple Lie algebra Pati-Salam model by Abdus Salam and Jogesh Pati also in 1974, who pioneered the idea to unify gauge interactions. The acronym GUT was first coined in 1978 by CERN researchers John Ellis, Andrzej Buras, Mary K. Gaillard, and Dimitri Nanopoulos, however in the final version of their paper they opted for the less anatomical GUM (Grand Unification Mass). Nanopoulos later that year was the first to use the acronym in a paper.

The year 1974 was the moment the idea crystallized. Howard Georgi and Sheldon Glashow at Harvard showed that all the particles of one generation could be seated inside a single mathematical object, the group SU(5), with electric charge falling out as a consequence rather than an input. Independently, Jogesh Pati and Abdus Salam reached for a different structure that braided quarks and leptons together even more intimately. Both proposals shared one radical wager: that the messy proliferation of particles and forces seen at low energy is a mask, and that behind it the universe is startlingly simple.

The fact that the electric charges of electrons and protons seem to cancel each other exactly to extreme precision is essential for the existence of the macroscopic world as we know it, but this important property of elementary particles is not explained in the Standard Model of particle physics.

While the description of strong and weak interactions within the Standard Model is based on gauge symmetries governed by the simple symmetry groups SU(3) and SU(2) which allow only discrete charges, the remaining component, the weak hypercharge interaction is described by an abelian symmetry U(1) which in principle allows for arbitrary charge assignments.

The observed charge quantization, namely the postulation that all known elementary particles carry electric charges which are exact multiples of one-third of the "elementary" charge, has led to the idea that hypercharge interactions and possibly the strong and weak interactions might be embedded in one Grand Unified interaction described by a single, larger simple symmetry group containing the Standard Model. This would automatically predict the quantized nature and values of all elementary particle charges.

Since this also results in a prediction for the relative strengths of the fundamental interactions which we observe, in particular, the weak mixing angle, grand unification ideally reduces the number of independent input parameters but is also constrained by observations. Grand unification is reminiscent of the unification of electric and magnetic forces by Maxwell's field theory of electromagnetism in the 19th century, but its physical implications and mathematical structure are qualitatively different. SU(5) is the simplest GUT.

The smallest simple Lie group which contains the Standard Model, and upon which the first Grand Unified Theory was based, is $SU(5) \supset SU(3) \times SU(2) \times U(1)$. Such group symmetries allow the reinterpretation of several known particles, including the photon, W and Z bosons, and gluon, as different states of a single particle field. However, it is not obvious that the simplest possible choices for the extended "Grand Unified" symmetry should yield the correct inventory of elementary particles.

The fact that all currently known matter particles fit perfectly into three copies of the smallest group representations of SU(5) and immediately carry the correct observed charges, is one of the first and most important reasons why people believe that a Grand Unified Theory might actually be realized in nature. The two smallest irreducible representations of SU(5) are 5 (the defining representation) and 10.

The elegance of the SU(5) construction was hard to overstate. Every known particle of a generation slotted into two tables -- a five-seat table for some quarks and the electron, a ten-seat table for the rest -- with electric charges that had to balance across each table for the theory to make mathematical sense. Charges that were mysterious inputs in the Standard Model became mandatory bookkeeping in the unified one. Physicists had spent half a century measuring why charge comes only in whole units; suddenly there was a reason.

This scheme has to be replicated for each of the three known generations of matter. It is notable that the theory is anomaly free with this matter content.

The hypothetical right-handed neutrinos are a singlet of SU(5), which means its mass is not forbidden by any symmetry; it doesn't need a spontaneous electroweak symmetry breaking which explains why its mass would be heavy (see seesaw mechanism).

This is already the largest simple group that achieves the unification of matter in a scheme involving only the already known matter particles (apart from the Higgs sector).

Since different Standard Model fermions are grouped together in larger representations, GUTs specifically predict relations among the fermion masses, such as between the electron and the down quark, the muon and the strange quark, and the tau lepton and the bottom quark for SU(5) and SO(10). Some of these mass relations hold approximately, but most don't (see Georgi-Jarlskog mass relation).

While GUTs might be expected to offer simplicity over the complications present in the Standard Model, realistic models remain complicated because they need to introduce additional fields and interactions, or even additional dimensions of space, in order to reproduce observed fermion masses and mixing angles. This difficulty, in turn, may be related to the existence of family symmetries beyond the conventional GUT models.

Due to this and the lack of any observed effect of grand unification so far, there is no generally accepted GUT model.

The collapse of minimal SU(5) taught a method as much as a lesson. Theorists responded not by abandoning unification but by enlarging the group: SO(10) absorbed the right-handed neutrino, supersymmetry re-tuned the running of the couplings, flipped models rearranged the assignments. Each escape hatch preserved the core idea while changing its predictions -- a normal day in science, though the accounting is brutal: half a century later, every surviving version remains unconfirmed.

The initial enthusiasm met a hard experimental wall. The simplest version of the theory predicted that protons -- the supposedly eternal building blocks of every atom -- should occasionally decay, and physicists calculated roughly how often. Huge underground detectors filled with water and patience waited years for a telltale flash that never came. The proton refused to disappear, and the simplest SU(5) model, so beautiful on paper, was ruled out. The lesson was double-edged: grand unification had produced a testable, falsifiable prediction, which is exactly what good physics is supposed to do -- and the experiment had said no.